Roll No. ....

TEE–601

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

POWER ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Why does unequal sharing takes place among series connected SCRs during steady state and dynamic state ? Draw the static and dynamic equalizing circuits for two series connected SCRs.

10 Marks (CO1, CO2)

OR

(b) Explain the string efficiency for series and parallel thyristors and also calculate the number of thyristors, each with a rating of 500 V, 75 A required in each branch of a series and parallel combination of a circuit with the total voltage and current rating of 7.5 kV and I kA. Assume derating factor of 12%. 10 Marks (CO1, CO2)

2. (a) Explain the operating principle of single phase half wave converter for purely inductive load with suitable diagram and waveforms.

10 Marks (CO2)

P. T. O.

OR

(b) A single phase full wave rectifier with center tapped transformer has a purely resistive load R. Determine : 10 Marks (CO2)

(i) Efficiency

(ii) Ripple Factor

(iii) Transformer Utilization Factor

(iv) Peak Inverse Voltage of Diode

3. (a) Describe the working of a single phase fully controlled bridge converter in the rectifying and inversion mode, with the help of relevant waveform. Derive the expressions for average output voltage and rms output voltage. 10 Marks (CO1, CO2)

OR

(b) Describe the various methods for turn on the thyristor.

10 Marks (CO1, CO2)

4. (a) Draw the circuit symbol and V-I characteristics of a SCR and explain its working with its mode of operations. 10 Marks (CO1, CO2)

OR

(b) A single phase full converter connected to 230 V, 50 Hz source is feeding a load R = 10 ohms in series with a large inductance that makes the load current ripple free. For a firing angle of 45 degree calculate the active power, reactive power, THD, CDF. 10 Marks (CO1, CO2)

5. (a) Draw the dynamic characteristics of SCR during its turn on process.

10 Marks (CO1)

OR

(b) Explain the significance of latching and holding current.

A dc supply of 100 V feeds a load resistance of 10 Ω and inductance of 5 H through a thyristor. The latching current of thyristor is 50 mA. Find the minimum width of gate pulse. 10 Marks (CO1)

50

Roll No. ....

TEE–602

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

POWER SYSTEM–II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you mean by per unit representation of power system quantities ? Derive impedance (in p.u) of the machine in terms of base MVA and base kV. 10 Marks (CO1)

OR

(b) A 5 kVA 400/200 V, 50 Hz single phase transformer has primary and secondary leakage reactance each of 2.5 Ω. Determine total reactance in p.u. 10 Marks (CO1)

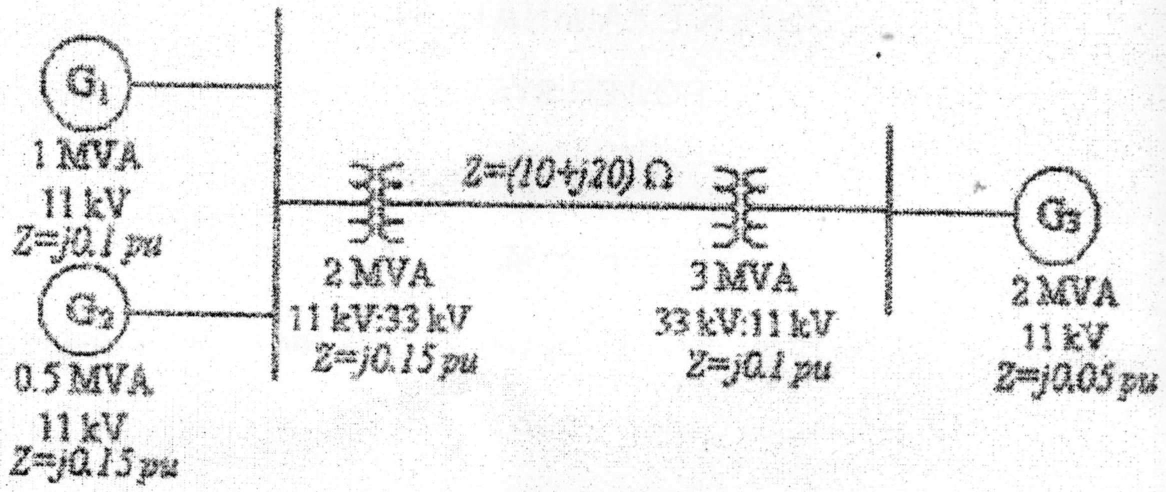
2. (a) Discuss Fortescue's theorem for the analysis of three-phase unbalanced system. 10 Marks (CO2)

P. T. O.

OR

(b) The one-line diagram for three generator system is shown in Fig. 1. Draw reactance diagram. Assume base values of voltage and MVA for low voltage side of 2 MVA transformer are 3 MVA and 11 kV.

10 Marks (CO2)



3. (a) Derive the complex power in terms of symmetrical components.

10 Marks (CO2)

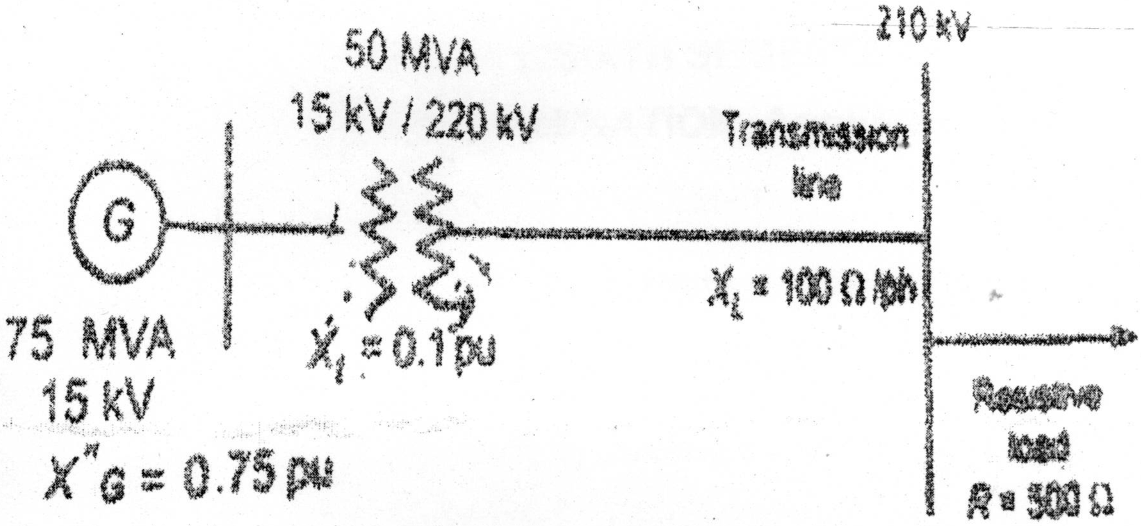
OR

(b) A balance star connected load takes 150 A from a balanced 3-phase 4-wire supply. If the fuses in two of the lines are removed, find the symmetrical components of the line current before and after the fuses are removed. 10 Marks (CO2)

4. (a) Apply symmetrical components analysis and interconnect the sequence networks in case of LG faults. 10 Marks (CO3)

OR

(b) For the system, as shown in figure given below, calculate the system current in p.u. 10 Marks (CO3)



5. (a) Give the general procedure used in the analysis of various types of shunt faults. Also, draw an equivalent network showing the interconnection of sequence networks to simulate LLLG fault. 10 Marks (CO3/CO1)

OR

(b) A three phase synchronous generator delivers 10 MVA at a voltage of 10.5 kV. The line impedance is 5Ω. Determine the voltage drop in the line in p.u. and in volts. Use the reference base as 12 MVA at 11 kV.

10 Marks (CO3/CO1)

50

Roll No. ....

TEE–603

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

POWER SYSTEM PROTECTION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) On the occurrence of a fault, how is it cleared by the protective devices ? Draw a suitable diagram to support the answer. 10 Marks (CO1)

OR

(b) Define Switchgear. Describe the essential features of a protective relay with reference to reliability and selectivity. 10 Marks (CO1)

2. (a) Analyze the behavior of a transmission line under sudden occurrence of fault and discuss doubling effect. 10 Marks (CO1)

OR

(b) How are relays classified on the basis to the functions performed ?

10 Marks (CO1)

3. (a) Describe the construction and working of a Buchholz relay.

10 Marks (CO2)

P. T. O.

OR

(b) Why is the reset to pick-up ratio low in dc attracted armature type relay as compared to ac relay ? Explain with reasons why an unpleasant noise (relay chattering) sometimes occurs in an ac attracted armature type relay. Discuss the remedial measures adopted to minimize this unpleasant noise. 10 Marks (CO2)

4. (a) Define the terms Plug Setting Multiplier and Time Multiplier setting as used in the context of an overcurrent relay. 10 Marks (CO2)

OR

(b) The current setting is set at 150%, TMS = 0.4; C.T. Ratio = 400/5, fault current is 6000 A. Determine the operating time of the relay. At TMS = 1 the operating time at various PSMs are shown in the table below : 10 Marks (CO2)

PSM	**Operating time in secs.**
2	10
4	5
5	4
8	3
10	2.8
20	2.4

5. (a) Describe the principle of impedance type distance relays. 10 Marks (CO3)

OR

(b) What is the principle of a differential relay ? What are its drawbacks and how are they overcome ? Explain. 10 Marks (CO3)

50

Roll No. ....

TOE–604

B. TECH. (ELECTRICAL ENGINEERING)

(SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

COMMUNICATION ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the block diagram of communication system.

10 Marks (CO1)

OR

(b) Define modulation. Explain the need of modulation in communication system.

10 Marks (CO1)

2. (a) Define frequency modulation. Explain Carson's rule for bandwidth of frequency modulation.

10 Marks (CO2)

OR

(b) What is the modulation index of an FM signal having a carrier swing of 100 kHz when the modulating signal has a frequency of 8 kHz.

10 Marks (CO2)

P. T. O.

3. (a) Explain any *one* method of generation of an amplitude modulated signal. 10 Marks (CO1)

OR

(b) Determine the ratio of modulating power to the total power with 100% modulation. 10 Marks (CO1)

4. (a) List out the various types of frequency modulation technique. Discuss any *one* of them in brief. 10 Marks (CO2)

OR

(b) Define noise. Explain various types of noise in communication system.

10 Marks (CO2)

5. (a) Explain the working principle of super heterodyne receiver with its block diagram. 10 Marks (CO1 and CO2)

OR

(b) Explain the concept of Pre-Emphasis and De-Emphasis in FM Systems.

10 Marks (CO1 and CO2)

50

Roll No. ....

TEE–606

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you understand by RISC and these two with the help of examples. 10 Marks (CO1)

OR

(b) What are the various data types available in computer system ? Explain. 10 Marks (CO1)

2. (a) How the Integer addition and subtraction take place in the computer system ? Explain with the help of flow chart. Illustrate the hardware implementation. 10 Marks (CO1)

OR

(b) How Booth's algorithm is different from conventional Multiplication algorithm ? Explain with the help of flow chart. 10 Marks (CO1)

P. T. O.

3. (a) Differentiate single bus and multi bus organizations. Explain some benefits of multi bus organizations. 10 Marks (CO1)

OR

(b) What is multi bus organization ? Illustrate with the help of diagram. Write down the control sequence for execution of instruction. Add R4,R5,R6 for 3 bus organization. 10 Marks (CO1)

4. (a) Draw and explain the instructions execution cycle with the help of an example. 10 Marks (CO1, CO2)

OR

(b) What is cache memory ? What are its various types ? How does the associative mapping take place ? 10 Marks (CO1, CO2)

5. (a) What do you mean by virtual memory and its need ? 10 Marks (CO2)

OR

(b) What do you understand by Memory Hierarchy ? Explain.

10 Marks (CO2)

50

Roll No. ....

TEE–610

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

INDUSTRIAL INSTRUMENTATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the working principle of a piezoelectric load cell in detail with proper diagrams. Mention the application of the same. 10 Marks (CO1)

OR

(b) Compare between elastic load cell and strain gauge load cell.

10 Marks (CO1)

2. (a) What do you understand by torque ? Explain the working of a Prony brake dynamometer along with proper diagram. Also, mention the various advantages of Prony brake dynamometer. 10 Marks (CO1)

OR

(b) What is an electric dynamometer ? What for an electric dynamometer is it used ? Explain eddy current type electric dynamometer in detail.

10 Marks (CO1)

P. T. O.

3. (a) What are the methods of vibration measurement ? Explain any *one* method in detail. 10 Marks (CO2)

OR

(b) Explain the mechanical type of vibration measuring instruments. Also, mention their advantages and limitations. 10 Marks (CO2)

4. (a) Explain the principle and operation of LVDT accelerometer. Mention its few applications. 10 Marks (CO2)

OR

(b) Give suitable comparison between a piezoelectric accelerometer and variable reluctance type accelerometer. 10 Marks (CO1)

5. (a) Explain the stroboscopic method of measuring speed. 10 Marks (CO1)

OR

(b) What are the various methods to measure speed ? Explain any *one* method in detail. 10 Marks (CO1)